

4.4 Entity Relationship Model (ER-Model)

ای آر ماڈل ڈیٹا بیس کا خاکہ ہے۔

CS XI

Chapter 4: Data and Analysis

Lecture 06 of 15

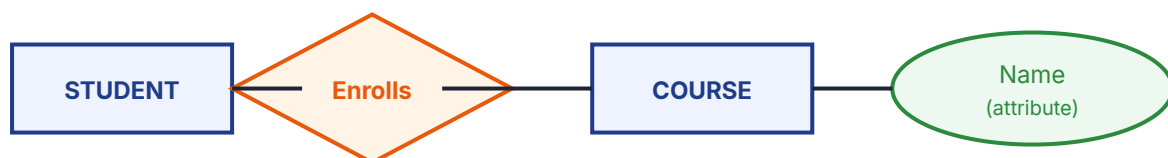
★ Golden

4.4 Entity Relationship Model (ER-Model) ★ Golden

LEARN IT

The **ER model** is the architect's blueprint: you draw what will be stored and how it connects *before* you create tables in Access. Three building blocks:

Component	Meaning	Symbol	Example
Entity	A real-world object we store data about	Rectangle	STUDENT, TEACHER, COURSE
Attribute	A property of an entity. PK is underlined	Oval	StudentID, Name, Class
Relationship	How two entities are connected	Diamond	Enrolls, Teaches, Has



Rectangle = entity, diamond = relationship, oval = attribute. Underline the primary-key attribute.

Cardinality (types of relationship)

Type	Meaning	Example
One-to-One (1:1)	One record $\leftrightarrow$ exactly one record. Rare; often for security.	One student has one ID card.
One-to-Many (1:M)	One record in A $\leftrightarrow$ many in B. The most common type.	One teacher teaches many students.
Many-to-Many (M:N)	Many in A $\leftrightarrow$ many in B.	Many students enrol in many classes.

COMMON MISTAKE

The M:N rule. A relational database cannot store M:N directly — it duplicates data. Break it into two 1:M relationships with a **junction table** (also called a bridge or associative table). Example: Enrollment(StudentID, ClassID, Date).

EXAM TIP

Always state cardinality on the diagram (1 and M at the ends of the line). For M:N, the long question is “how do you resolve it?” — draw the junction entity with two foreign keys.

اردو میں سمجھیں

ای آر ماڈل ڈیٹا بیس کا خاکہ ہے۔ مستطیل ہستی، بیضوی صفت، لوزی تعلق۔ تعلقات 1:1، 1:M یا M:N ہوتے ہیں۔ M:N کو دو 1:M میں توڑنے کے لیے جنکشن ٹیبل بنانا پڑتا ہے۔

CHECK YOURSELF

1. What symbol is used for a relationship?
2. A hospital: one patient, many appointments. Cardinality?

Answers: 1. Diamond. 2. 1:M (Patient 1 — has — M Appointment).

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. An ER model is drawn:

- a. Before building tables
- b. After deleting the DB
- c. Only in Python
- d. Only in OSI

MCQ 2. An entity is:

- a. A real-world thing we store
- b. A Python keyword
- c. A cable
- d. A sprint

MCQ 3. A relationship 1:M means:

- a. One A, many B
- b. Many A, many B only
- c. One to one only
- d. No link

MCQ 4. M:N in an RDBMS needs:

- a. A bridge/associative table
- b. A pie chart
- c. A stack
- d. A NAND

MCQ 5. A diamond in ER often shows:

- a. A relationship
- b. An attribute oval only
- c. A PK square
- d. A report

MCQ 6. Cardinality is:

- a. 1:1, 1:M, M:N
- b. Big O
- c. TCP window
- d. Variance

Extra short questions

1. Define ER model.
2. Three relationship types with examples.
3. How do you convert 1:M to tables?
4. Why ER first?

Extra long question

1. Draw ER for Library: Member, Book, Borrow. Mark keys, cardinality, and how you resolve M:N. ★ Golden.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Entity-Relationship: conceptual design of entities, attributes, relationships.

2. 1:1 person–NIC; 1:M mother–children / member–borrows; M:N students–courses.
3. PK of the 1 side copied as FK on the M side.
4. Cheaper to fix a diagram than a populated database.

Long

1. Member 1—M Borrow M—1 Book (Borrow as relationship-entity with date). If Book–Member M:N, Borrow is the bridge with composite/surrogate key.

Next lecture: 4.5 Referential Integrity (Cascade Update & Delete)